The Impact of Transboundary Haze on Lung Cancer Mortality in Thailand

A Policy Paper

By: Jirawat Poongam
MEP 23207
Advisor: Assistant Professor Jun Goto

September 2024

Two-year Master's Program of Public Policy
National Graduate Institute for Policy Studies
Tokyo, Japan

ABSTRACT

This study investigates the long-term health impacts of exposure to PM2.5 air pollutants on lung cancer mortality. Using data from ground-based stations, Aerosol Optical Depth (AOD), and estimated PM2.5 levels, we found a significant correlation between long-term PM2.5 exposure and increased lung cancer mortality. Specifically, over a 10-year period, especially during haze seasons, each one microgram per cubic meter (μg/m^3) increase in PM2.5 concentration is associated with a 0.275 percentage point rise in lung cancer mortality probability. While no conclusive evidence was found linking PM2.5 to lung cancer mortality over shorter exposure periods, such as five years, the long-term impacts underscore the urgent need for effective air quality management policies. Policymakers should consider enhanced air quality monitoring and regulation, public awareness campaigns, targeted health interventions, and support for further research to mitigate the adverse health effects of PM2.5 exposure.

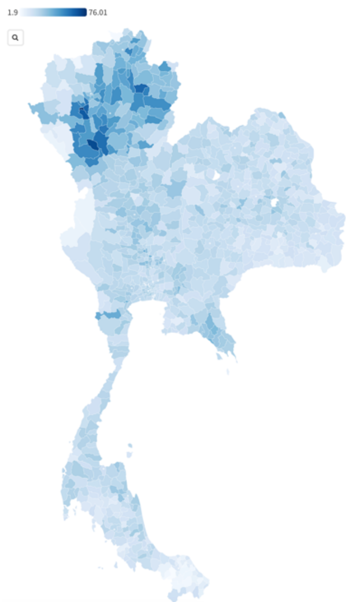
Keywords: Lung Cancer Mortality, Southeast Asia, Transboundary Haze, Air Quality, Particulate Matter, PM2.5, Aerosol Optical Depth, AOD, Satellite Imagery, Remote Sensing

CONTENTS

INTRODUCTION

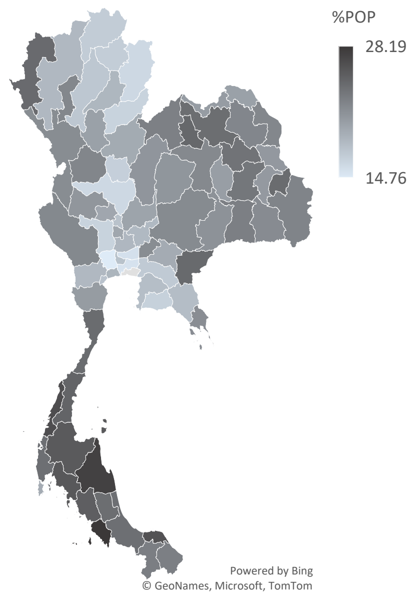
The heightened mortality rate from lung cancer in the northern region of Thailand, despite similar smoking behaviors to other regions, suggests the influence of additional risk factors. The data shown in Figure 1 indicates that on average the northern area of Thailand has a greater death rate from lung cancer than any other location from 2016 – 2020. But not from the smoking (Figure 2), because the ratio of smoking population in the north not different from another region. Therefore, this disparity could be attributed to various environmental hazards, such as exposure to indoor radon, smoke, and air pollution, which have been identified as significant contributors to lung cancer risk[1].

Figure 1: Average death rate from lung cancer per 100,000 persons in Thailand from 2016 to 2020



Source: Strategy and Planning Division, Ministry of Public Health, Thailand

Figure 2: Average tobacco consumption rates aged 15 years and older in Thailand from 2011 and 2017



Source: National Statistical Office of Thailand

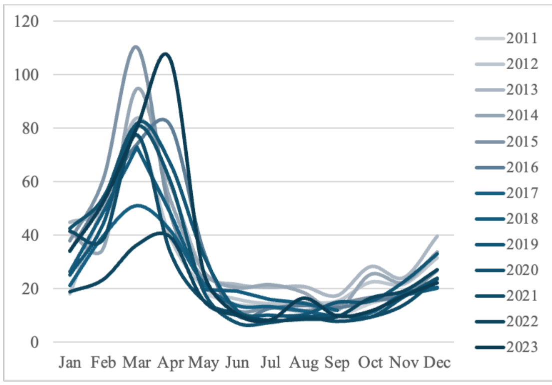
Southeast Asia has a long history of annually transboundary haze issues, which are mostly brought on by forest fires, deforestation, land clearing activities, and slash-and-burn methods—which are the cheapest and simplest ways to clear land for traditional agriculture. Large volumes of smoke and particulate matter are released into the atmosphere as a result of these activities, which have an impact on human health, activity, and economy. Small particles, such as PM10 or PM2.5[2],

[1] The American Cancer Society medical and editorial content team https://www.cancer.org/content/dam/CRC/PDF/Public/8704.00.pdf
[2] PM2.5 is particulate matter that is 2.5 micrometers or smaller in diameter.

pose a health risk because they can get deeply into the respiratory system, enter the circulation, reach the lungs, and entering the bloodstream, where they are associated with a higher risk of lung cancer.

Southeast Asia has two independent haze cycles annually: the first occurs in the northern parts of Thailand, Myanmar, and Laos from January to April, for example, the figure 3 shows the seasonal pattern of PM 2.5 in the center city of Chiang Mai, Thailand from ground-based air quality measurement stations. While the second appears in Indonesia, Malaysia, and Singapore from July to October. Relatively few studies have been conducted in the past on the first occurrence of haze-related issues from January to April, especially when it comes to quantifying the impact. In the latter cycle, there have been numerous studies on a variety of topics and a range of regional cooperative action through the Association of Southeast Asian Nations (ASEAN).

Figure 3: Monthly average PM 2.5 (µg/m3) in the center Chiang Mai, Thailand from ground-based air quality measurement stations (36T)



Source: Pollution Control Department, Thailand

Therefore, while smoking remains a primary risk factor for lung cancer, the unique environmental conditions and pollution levels in northern Thailand's atmosphere likely play a crucial role in the region's elevated lung cancer mortality rates. Understanding the impact of haze problems across Thailand, Myanmar, and Laos from January to April on the people experiencing air pollution problems is essential for developing targeted public health interventions and policies to mitigate these risks.

LITERATURE REVIEW

This section needed to be divided into three parts in order to review the literature associated with this research areas. Firstly, understanding particulate matter and its significance. Secondly, investigating studies linking exposure to PM2.5 and mortality due to lung cancer. Thirdly, exploring the shortcomings of ground-based particulate matter measurement stations and how aerosols address these issues.

What is particulate matter

The World Health Organization (WHO) first set goals related to clean air and air quality in its "Air Quality Guidelines for Europe" published in 1987. The guidelines covered a range of pollutants, including particulate matter (PM), sulfur dioxide (SO_2), nitrogen dioxide (NO_2), carbon monoxide (CO), and ozone (O_3). At that time, WHO primarily focused on PM without distinguishing between different sizes of particles (total suspended particles: TSP) but these initial guidelines laid the groundwork for subsequent updates and expansions.

Key contributions came from numerous studies and regulatory efforts, particularly in the United States, where the Environmental Protection Agency (EPA) played a significant role. In 1997, the EPA established the first air quality standards for PM2.5, influenced by extensive research. For example, Pope et al. (1991) demonstrated a significant correlation between daily PM10 levels and adverse respiratory health outcomes, highlighting the immediate impact of particulate pollution on human health. Dockery et al. (1993) study further underscored the risks by revealing a strong association between elevated levels of fine particulate matter and increased mortality rates from cardiovascular and respiratory diseases across six U.S. cities.

The World Health Organization (WHO) then first set specific guidelines for PM2.5 in their "Air Quality Guidelines" published in 2005. Recognizing the severe health impacts of fine particulate matter, the WHO recommended an annual mean concentration of PM2.5 not exceeding 10 $\mu g/m^3$ and a 24-hour mean concentration not exceeding 25 $\mu g/m^3$. The establishment of these targets marked a significant step in global public health efforts to reduce air pollution and its detrimental effects on human health.

The association between PM2.5 exposure and lung cancer mortality

Numerous studies have investigated the association between PM2.5 exposure and lung cancer mortality rates, revealing significant and consistent links. For example, Pope et al. (2002) analyzed data from the American Cancer Society's Cancer Prevention Study II, tracking over 500,000 adults and finding that long-term exposure to PM2.5 significantly increased lung cancer mortality, even after controlling for smoking and other confounders. Similarly, Beelen et al. (2014) conducted a large-scale study across several European countries, part of the ESCAPE project, which confirmed the positive association between PM2.5 and lung cancer mortality. Turner et al. (2011) focused on never-smokers within the same cohort, demonstrating that even non-smokers are at increased risk of lung cancer mortality due to PM2.5 exposure.

Haze, marked by elevated levels of fine particulate matter (PM2.5), presents substantial health risks. These fine particles can carry polycyclic aromatic hydrocarbons (PAHs), which are byproducts of incomplete combustion from sources like vehicle emissions, industrial activities, and biomass burning. When inhaled, PM2.5 particles penetrate deep into the respiratory system, delivering PAHs directly to lung tissues. The carcinogenic potential of PAHs is well-documented, as they can induce mutations and promote the development of cancerous cells. Consequently, long-term exposure to these particles increases the risk of respiratory illnesses and various forms of cancer. Loomis et al. (2013) highlighted the carcinogenic potential of outdoor air pollution, emphasizing PM2.5's role in elevating lung cancer risk.

The northern part of Southeast Asia faces a persistent issue with annual transboundary haze, primarily caused by forest fires, deforestation, land clearing activities, and slash-and-burn methods from January to April (the hot season). Various studies have investigated the health effects of this haze. For instance, Pongpiachan and Paowa (2015) found a significant increase in hospital visits and admissions for children under 15 years old in Chiang Mai during the haze season. Similarly, Trang and Tripathi (2014) identified a strong correlation between respiratory diseases and high PM10 levels in the northeastern districts of Chiang Mai in the second week of March. However, most research is limited to specific areas due to constraints such as data coverage, health records, and ground-based air measurement stations. Furthermore, there is a lack of research on the effects of seasonal haze in northern Southeast Asia on lung cancer mortality rates.

The limitations of PM2.5 measurement and their association with aerosols

Measuring small particles like PM2.5 faces challenges with ground-based stations in terms of coverage especially in developing countries. To address this, recent literature has utilized advancements in technology, using satellite imagery and remote sensing to estimate air quality and related data. These methods, such as aerosol optical depth (AOD) data from instruments like the Moderate Resolution Imaging Spectroradiometer[3] (MODIS). The use of AOD as a proxy for estimating PM2.5 levels gained prominence in the early 2000s. AOD quantifies how aerosol particles in the atmosphere diminish solar light, as observed from satellites. This

[3]Both Terra and Aqua satellite contain Moderate Resolution Imaging Spectroradiometer (MODIS) sensor that can measure the aerosol optical depth (AOD) of the atmosphere

approach proves beneficial for estimating spatial and temporal PM2.5 concentrations, particularly in regions lacking ground-based monitors.

One of the pioneering studies that linked AOD to PM2.5 was conducted by van Donkelaar et al. (2010). In their research, they developed a method to derive global estimates of PM2.5 from satellite AOD measurements by applying a combination of aerosol transport models and satellite observations. Boys et al. (2014) utilized a 15-year global time series of satellite-derived AOD to estimate PM2.5 concentrations, finding robust associations with increased lung cancer mortality. Similarly, Li et al. (2017) focused on China, using satellite AOD data to reveal significant links between PM2.5 exposure and lung cancer deaths. Liu et al. (2018) validated the use of MODIS satellite data for estimating PM2.5 levels in Beijing, also demonstrating a significant positive association with lung cancer mortality. These studies underscore the reliability and importance of AOD in monitoring air pollution and assessing its health impacts, especially in regions where traditional ground-based measurements are not feasible.

Recently, numerous studies have utilized Global (GL) Annual PM2.5 Grids from MODIS, MISR, and SeaWiFS Aerosol Optical Depth (AOD) data, published by Hammer et al. (2020), Hammer et al. (2022) and SEDAC, as a predicted PM2.5 levels and examine their relationship with lung cancer. For example, Guo et al. (2020) found that predicted PM2.5 levels had significantly adverse effects on the incidence rates of lung cancer for both males and females in China.

In conclusion, previous studies have demonstrated a significant link between air pollution, particularly PM2.5, and human health. Consequently, improving air quality is a crucial aspect of global public health initiatives aimed at mitigating its harmful effects. However, there has been limited research on the impact of seasonal haze in the northern part of Southeast Asia during the hot season. Studies in this area are often restricted by data coverage limitations, such as health records and ground-based air measurement stations. Although recent research has utilized technological advancements like AOD data from MODIS, most of these studies have focused on the correlation between AOD and PM2.5 without linking it to lung cancer mortality. Additionally, predicted PM2.5 from Global (GL) Annual PM2.5 Grids from MODIS, MISR, and SeaWiFS Aerosol Optical Depth (AOD) cannot identify the haze season due to its reliance on annual averages. Thus, a research gap exists in understanding the relationship between transboundary haze's impacts and lung cancer mortality in northern Southeast Asia.

METHODOLOGY

This study employs a statistical analysis approach to investigate the association between PM2.5 exposure and lung cancer mortality in Southeast Asia. We adopted methodologies from previous studies, integrating satellite-derived data and ground-based monitoring data to assess PM2.5 concentrations and their health impacts. The primary steps in our methodology include data collection and processing, and health impact analysis.

Data collection and processing

Data from three main sources were used to examine the effects of transboundary haze in northern Southeast Asia, including Myanmar, Laos, and Thailand: lung cancer mortality, data from ground-based air quality measuring stations, and satellite data. However, there are many limitations including data availability. Lung cancer mortality rate at the provincial or district level available only in Thailand, thus the boundary of this research limit on district level of Thailand only.

Table 1: Descriptive statistics

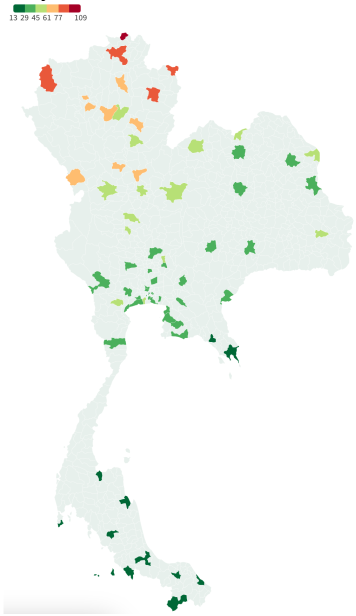
Variables	Region	Obs	Mean	SD
Lung Cancer Mortality (per 100,000 people)	TH	4,640	22.28	11.11
	Central	1,295	23.35	8.50
	North	980	30.67	15.03
	Northeast	1,610	18.88	7.47
	South	755	16.79	8.68
Particulate Matter 2.5 (µg/m3)	TH	4,897	23.76	16.13
	Central	2,621	23.92	12.17
	North	1,195	27.48	24.09
	Northeast	476	24.64	15.47
	South	605	15.05	5.12
Aerosol Optical Depth	TH	259,727	404.52	192.99
	Central	67,891	429.45	165.34
	North	54,962	443.30	228.65
	Northeast	90,115	434.63	196.13
	South	46,759	264.72	87.43
Tobacco Consumption Rates (percent)	TH	154	20.92	3.76
	Central	52	18.88	3.13
	North	34	19.27	3.48
	Northeast	40	22.25	2.58
	South	28	24.84	2.68
Household income (Thai baht)	TH	770	20,423.44	6,987.84
	Central	260	24,207.62	7,449.38
	North	170	16,775.54	4,446.61
	Northeast	200	16,824.63	4,707.09
	South	140	22,966.43	6,540.24
Elevation (meter)	TH	928	190.48	194.49
	Central	259	61.17	99.20
	North	196	396.59	273.51
	Northeast	322	206.38	94.27
	South	151	110.84	96.28

Note: This table presents various health and environmental statistics across four regions: Central, North, Northeast, and South. For each variable, the table lists the number of observations (Obs), mean values (Mean), and standard deviations (SD).
Source: Ministry of Public Health, National Statistical Office, Pollution Control Department in Thailand, and the United States Geological Survey (USGS) NASA

Table 1 provides descriptive information on all relevant data used in this research. The details are as follows. Lung cancer mortality in Thailand from 2017 to 2021 in 928 districts, as published by the Strategy and Planning Division of the Ministry of Public Health. The data shows that the average death rate from lung cancer in northern region is 30.67 per 100,000 people, higher than in any other regions.

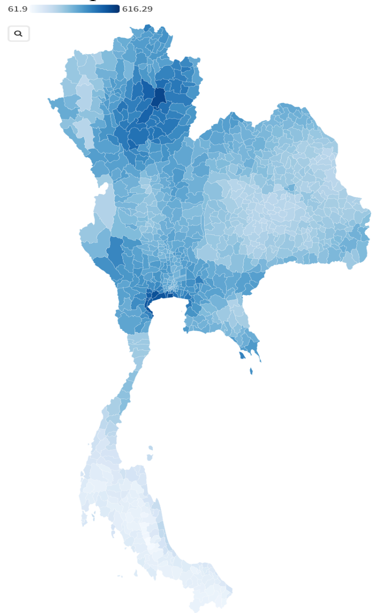
PM2.5 were measured from ground-based air measurement stations published by Pollution Control Department of Thailand. Table 1 demonstrates that the northern region has the highest average PM2.5 concentration among other regions, with 27.48 micrograms per cubic meter of air, or µg/m3. Nevertheless, just 61 stations—or 6.6% of Thailand's total of 928 districts—are covered, and some of them have a maximum historical data period of 10 years[4]. The locations of Thailand's ground-based air quality monitoring stations are shown in Figure 4. As a result, stations are restricted to some locations, which means that the data they provide could not fully reflect wider regions or the variability of air quality over time. To overcome the second limitation, the recent technological advancements, air quality and related data can be estimated though satellite images and remote sensing without the limitations of data coverage such as Aerosol Optical Depth (AOD) as demonstrate in Figure 5.

Figure 4: Average PM2.5 from Ground-based air quality measurement stations in Thailand



Source: Pollution Control Department, Thailand

Figure 5: The monthly average Aerosol Optical Depth from MODIS in Feb 2021.



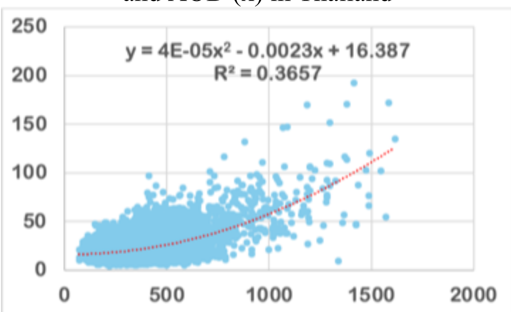
Source: NASA LP DAAC at the USGS EROS Center

[4] The first publicly accessible data set dates back to 2011, when there were 61 stations. Up until 2021, the number of stations remained quite stable. By 2023, there will be 96 stations, up from the 77 stations in 2021.

The NASA Earth Observing System Data and Information System (EOSDIS) provided the AOD data from the Moderate Resolution Imaging Spectroradiometer (MODIS)[5]. Table 1 illustrates the monthly average AOD for each of Thailand's 928 districts from January 2001 to April 2024, retrieved from Google Earth Engine. They only reported on a scale that ranged from -100 to 8000 at the highest level, not in units of AOD. The average AOD in the northern region was 443.30, the highest of all the regions.

AOD and PM2.5 concentrations have been found to positively correlate in the literature review section, indicating that as AOD increases, so do the concentrations of these fine particulate matters. Figure 6 in the instance of Thailand similarly demonstrates a positive relationship. However, AOD measures how much sunlight is blocked by aerosols, which are tiny particles suspended in the atmosphere. Aerosols might not directly affect human health in the same way as PM 2.5 does. Thus, the goal of this study is to investigate how AOD and PM2.5 relate to lung cancer.

Figure 6: Relationship between monthly average PM 2.5 (y) and AOD (x) in Thailand



Source: Author's calculation

To address the issue of confounding variables in a study examining the relationship between lung cancer mortality, PM2.5, and AOD, it is important to control for variables that might influence both the exposure (PM2.5 and AOD) and the outcome (lung cancer mortality). In this case, tobacco consumption rate, household income and elevation are the potential confounders. First, the Thailand's tobacco consumption rate is defined as the proportion of smokers in each province who were at least 15 years old, based on the surveys carried out by the National Statistics Office in 2011 and 2018. According to Table 1, the average percentage of

[5] Aerosol Optical Depth (AOD) is a crucial measure of the transparency of the atmosphere, indicating the presence of particles such as dust, smoke, and other pollutants. The MODIS (Moderate Resolution Imaging Spectroradiometer) instruments aboard NASA's Terra and Aqua satellites provide valuable data on AOD. These instruments capture daily global AOD measurements at a resolution of 1 km, which are essential for understanding air quality and climate change.

smokers in the northern areas is 19.27 percent, which is less than the national average of 20.92 percent.

Second, household income surveyed by the National Statistics Office in each province every two years between 2004 and 2021 (10 times). Table 1 indicates that the average household income in the northern region is 16,775.54 Thai baht, which is less than the national average. Third, digital elevation data from The Shuttle Radar Topography Mission (SRTM) was used to provide geographic information in 2000, such as elevation in each district. NASA JPL contributed this SRTM V3 product (SRTM Plus), which was obtained via Google Earth Engine. Table 1 data shows that the average elevation in the northern region is 396.59 meter above sea level which is higher than in other regions.

Health impact analysis

This study examines the long-term exposure to air pollutants such as PM2.5 and AOD, which has been linked to an increased risk of developing lung cancer. A cumulative exposure assessment approach was used, where long-term exposure to air pollutants is typically assessed by averaging pollutant levels over extended periods—in this study, over 5, 10, and 20 years. The analysis is represented by the following equation:

$$Lung\ cancer_d = \beta_0 + \beta_1 Air\ Quality_{normal_d} + \beta_2 Air\ Quality_{haze_d} + \gamma Controls_d + \varepsilon_d \quad (1)$$

Where $Lung\ cancer_d$ is average lung cancer mortality rate from 2017 to 2021 in district d. $Air\ Quality\_normal_d$ is average of air quality outside haze season in district d from May to December over 5, 10, and 20 years of PM2.5, AOD, and estimated PM2.5 from the first equation. $Air\ Quality\_haze_d$ is differentiate between the average air quality during haze season (January to April) and outside the haze season (May to December) in district d over 5, 10, and 20 years of PM2.5, AOD, and estimated PM2.5 from the first equation. $Controls_d$ are control variables such as tobacco consumption rate in each province, household income in each province, as well as additional air quality factors like ozone, carbon monoxide, sulfur dioxide, and nitrogen dioxide. The newly prepared data for the health impact analysis is shown in Table 2.

Table 2: Descriptive statistics for health impact analysis

Variable		Obs	Mean	Std. dev.	Min	Max
Average lung cancer mortality rate from 2017 to 2021		928	22.28	8.91	1.898	76.01
Average tobacco consumption rate in 2011 and 2018		928	20.94	3.22	14.755	28.19
Average household income (THB) from 2004 to 2021		928	20,687.87	6,620.09	10,844.97	41,698.14
Carbon monoxide (CO) mol/m^2 from 2019 to 2021		928	0.0395	0.0034	0.0296	0.0449
Nitrogen dioxide (NO2) mol/m^2 from 2019 to 2021		928	0.0000639	0.0000231	0.0000405	0.0001635
Ozone (O3) mol/m^2 from 2019 to 2021		928	0.1175	0.0005	0.1158	0.1182
Sulfur dioxide (SO2) mol/m^2 from 2019 to 2021		928	-8.29E-06	0.0000113	-0.0000367	0.0000335
5 years annual average air quality (X) from 2017 - 2021						
PM2.5	All year average (Jan-Dec)	65	22.41	5.81	9.15	38.69
	During haze seasons (Jan-Apr)	61	36.36	13.94	10.67	87.06
AOD	All year average (Jan-Dec)	928	396.04	67.11	212.06	538.26
	During haze seasons (Jan-Apr)	928	501.86	125.10	202.28	771.74
Estimated PM2.5	All year average (Jan-Dec)	928	23.38	5.39	8.81	36.15
	During haze seasons (Jan-Apr)	928	35.70	8.09	14.95	70.88
10 years annual average air quality (X) from 2011 - 2021						
PM2.5	All year average (Jan-Dec)	65	22.63	6.04	9.15	38.69
	During haze seasons (Jan-Apr)	61	36.86	14.09	10.67	87.06
AOD	All year average (Jan-Dec)	928	413.72	71.58	228.83	553.78
	During haze seasons (Jan-Apr)	928	523.96	129.00	207.36	780.85
Estimated PM2.5	All year average (Jan-Dec)	928	24.03	5.54	8.74	35.56
	During haze seasons (Jan-Apr)	928	36.71	8.17	16.58	64.59
20 years annual average air quality (X) from 2001 - 2021						
AOD	All year average (Jan-Dec)	928	410.57	75.37	221.44	580.34
	During haze seasons (Jan-Apr)	928	505.62	128.06	193.06	762.91
Estimated PM2.5	All year average (Jan-Dec)	928	23.80	5.55	8.63	35.70
	During haze seasons (Jan-Apr)	928	35.78	7.85	16.39	60.30

Note: The number of observations for each variable is indicated in the "Obs" column. Average lung cancer mortality rate from 2017 to 2021 published by the Strategy and Planning Division of the Ministry of Public Health, which is 22.28 per 100,000 people, and the average tobacco consumption rate in 2011 and 2018 surveyed by National Statistical Office, which is 20.94 percent of smokers. The average household income from 2004 to 2021 surveyed by National Statistical Office is 20,687.87 Thai baht. Pollutant data from 2019 to 2021 from European Space Agency reveal the following mean concentrations: carbon monoxide (CO) at 0.0395 mol/m², nitrogen dioxide (NO2) at 0.0000639 mol/m², ozone (O3) at 0.000154 mol/m², and sulfur dioxide (SO2) at -8.92E-06 mol/m². The air quality data is detailed for three periods: 5 years (2017-2021), 10 years (2011-2021), and 20 years (2001-2021). It includes measurements of PM2.5 published by Pollution Control Department, Aerosol Optical Depth (AOD) from NASA, and the estimated PM2.5 levels for the entire year and specifically during haze seasons (January to April).
Source: Ministry of Public Health, National Statistical Office, Pollution Control Department in Thailand, the United States Geological Survey (USGS) NASA, and European Space Agency (ESA)

RESULTS AND DISCUSSION

The results of the health impact analysis from long-term exposure to air pollutants are divided into three parts based on the independent variables: PM2.5 from ground-based stations, AOD, and estimated PM2.5[6]. This study found the linkage between long-term PM2.5 exposure and lung cancer mortality especially during haze season. The results of AOD, which covered broader areas and a longer historical data period than PM2.5 and was also linked to lung cancer mortality, support the findings from ground-based air measuring stations.

The impact of PM2.5 from ground-based stations on lung cancer mortality

On average PM2.5 correlated with the mortality rate from lung cancer for 10 years exposure windows. For every one microgram increase in PM2.5, or µg/m3, the probability of lung cancer mortality rate increases by 0.275 percentage point. These effects especially came from during haze season as shown in the fourth regression model. However, we cannot conclude the linked of PM2.5 with lung cancer mortality rate in short-term or 5 years exposure in case of Thailand.

Table 3: The impact of PM2.5 from ground stations for 10- and 5-years exposure on lung cancer mortality

Y = Average Lung cancer Mortality Rate (2017-2021)	Exposure PM2.5 on average 10 Years (2011-2021)				Exposure PM2.5 on average 5 Years (2017-2021)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
PM2.5 (Jan-Dec)	0.392***	0.275*			0.427***	0.214		
	-0.125	-0.138			-0.134	-0.181		
PM2.5 outside haze season (May-Dec)			0.331**	0.116			0.378**	0.067
			-0.139	-0.147			-0.156	-0.191
PM2.5 Haze (During Jan-Apr different from May-Dec)			0.175**	0.186**			0.180**	0.104
			-0.076	-0.084			-0.078	-0.111
Control variables								
Tobacco consumption rates		✓		✓		✓		✓
Household income		✓		✓		✓		✓
Air pollutants						✓		✓
Observations	65	65	61	61	65	65	61	61
R-squared	0.142	0.183	0.175	0.233	0.156	0.273	0.184	0.31

Note: The analysis covers two periods: a 10-year average (2011-2021) and a 5-year average (2017-2021). Columns (1) and (2) focus on the 10-year average exposure to PM2.5 for the full year (Jan-Dec), while columns (5) and (6) focus on the 5-year average exposure to PM2.5 for the full year (Jan-Dec). Columns (3) and (4) examine the 10-year average PM2.5 concentration during the non-haze season (May-Dec) and the haze season (Jan-Apr), respectively, while columns (7) and (8) do the same for the 5-year average.
Source: Author's calculation

The impact of aerosol optical depth (AOD) on lung cancer mortality

AOD cannot be directly interpreted as an indicator of air quality since aerosols exist in the atmosphere and may not have the same direct health effects on humans as PM2.5. However, compared to ground-based air quality measuring

[6] The results of PM2.5 estimation and its impact on lung cancer mortality provided in Appendix A.

stations, using AOD has the advantage of covering more districts and having longer historical data, and several prior research studies found a correlation between AOD and PM2.5. According to this study, there was a correlation between the mortality rate from lung cancer and the AOD for exposure periods of 20 years. The likelihood of a lung cancer death rate rises by 0.015 percentage points for every unit increase in AOD. However, according to the twelfth, sixteenth, and twentieth regression models, the season of haze was the source of the positive correlation between AOD and lung cancer mortality. The findings from PM2.5 exposure is supported by these AOD exposure results.

Table 4: The impact of AOD for 20- 10- and 5-years exposure on lung cancer mortality

Y = Average Lung cancer Mortality Rate (2017-2021)	Exposure AOD on average 20 Years (2001-2021)				Exposure AOD on average 10 Years (2011-2021)				Exposure AOD on average 5 Years (2017-2021)			
	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
AOD (Jan-Dec)	0.036***	0.015***			0.038***	0.008			0.041***	0.003		
	-0.003	-0.004			-0.003	-0.005			-0.004	-0.011		
AOD outside haze season (May-Dec)			-0.019***	-0.024***			-0.014**	-0.030***			-0.016**	-0.018
			-0.006	-0.006			-0.007	-0.007			-0.007	-0.011
AOD Haze (During Jan-Apr different from May-Dec)			0.052***	0.047***			0.043***	0.036***			0.044***	0.036***
			-0.004	-0.004			-0.004	-0.005			-0.004	-0.007
Control variables												
Tobacco consumption rate		✓		✓		✓		✓		✓		✓
Household income		✓		✓		✓		✓		✓		✓
Air pollutants										✓		✓
Observations	928	928	928	928	928	928	928	928	928	928	928	928
R-squared	0.093	0.148	0.221	0.245	0.091	0.165	0.175	0.225	0.096	0.225	0.194	0.255

Note: The analysis covers three periods: a 20-year average (2001-2021), a 10-year average (2011-2021) and a 5-year average (2017-2021). Columns (9) and (10) focus on the 20-year average exposure to PM2.5 for the full year (Jan-Dec), columns (13) and (14) focus on the 10-year average, and columns (17) and (18) focus on the 5-year average. Columns (11) and (12) examine the 20-year average PM2.5 concentration during the non-haze season (May-Dec) and the haze season (Jan-Apr), columns (15) and (16) do the same for the 10-year average, and columns (19) and (20) for the 5-year average. Control variables include tobacco consumption rates, household income, and other air pollutants. Each column presents coefficient estimates, standard errors, number of observations, and R-squared values.
Source: Author's calculation

Discussion

This study highlights a significant link between long-term PM2.5 exposure and increased lung cancer mortality, particularly during haze seasons. The alignment between AOD-based PM2.5 estimates and ground-based measurements, shown in Figures A2 and A4, further supports our findings. While our results underscore the severe health risks associated with long-term exposure, we found no conclusive evidence linking PM2.5 to lung cancer mortality over shorter periods, such as five years, in Thailand. This may be due to the time required for cancer to develop, suggesting the need for further research on shorter-term exposures.

Future research should explore additional dimensions of PM2.5 exposure, including the use of rolling windows for analysis to capture variations over different time frames. Expanding research to include other regions and incorporating diverse methodologies will provide a more comprehensive understanding of how varying exposure durations affect health. Enhanced data collection and longitudinal studies will be crucial for refining policy measures and improving public health strategies.

CONCLUSION AND POLICY IMPLICATION

The health impact analysis demonstrates a significant correlation between long-term exposure to air pollutants, particularly PM2.5, and lung cancer mortality. This correlation is observed using PM2.5 data from ground-based stations, and AOD. The study highlights that PM2.5 exposure over a 10-year period, especially during haze seasons, increases the probability of lung cancer mortality by 0.275 percentage points for each microgram per cubic meter ($\mu g/m^3$) increase in PM2.5 concentration. While the analysis confirms a strong association over a decade-long exposure, it does not find conclusive evidence for a link between PM2.5 and lung cancer mortality over shorter exposure periods, such as five years, in the context of Thailand.

The findings of this study highlight the urgent need for effective air quality management policies to reduce the adverse health effects of long-term PM2.5 exposure. The clear link between PM2.5 levels and lung cancer mortality, especially during haze seasons, underscores the importance of stringent regulatory measures to control air pollution.

Policymakers should consider several critical actions to address this public health issue. First, there should be enhanced air quality monitoring and regulation, particularly in urban and industrial areas. Second, public awareness about the health risks of PM2.5 exposure needs to be increased to encourage the adoption of practices that reduce pollution. Third, targeted health interventions should be developed for populations at higher risk and vulnerable groups. Fourth, specific action plans for haze seasons should be implemented to reduce activities that contribute to PM2.5 emissions. Finally, further research should be supported through better data collection and funding to understand the long-term health impacts of PM2.5 exposure.

While the study does not find conclusive evidence linking PM2.5 to lung cancer mortality over shorter exposure periods, such as five years, in Thailand, the long-term impacts are evident and necessitate immediate action. By implementing these policy recommendations, governments can significantly reduce the health burden of air pollution and improve public health outcomes.

REFERENCES

Beelen, R., Hoek, G., van den Brandt, P. A., Goldbohm, R. A., Fischer, P., Schouten, L. J., ... & Brunekreef, B. (2014). Effects of long-term exposure to air pollution on natural-cause mortality: An analysis of 22 European cohorts within the multicentre ESCAPE project. The Lancet, 383(9919), 785-795.

Boys, B. L., Martin, R. V., van Donkelaar, A., MacDonell, R. J., Hsu, N. C., Cooper, M. J., ... & Burnett, R. T. (2014). Fifteen-year global time series of satellite-derived fine particulate matter. Environmental Health Perspectives, 122(10), 989-996.

Dockery, D. W., Pope, C. A., III, Xu, X., Spengler, J. D., Ware, J. H., Fay, M. E., Ferris, B. G., Jr., & Speizer, F. E. (1993). An association between air pollution and mortality in six U.S. cities. New England Journal of Medicine, 329(24), 1753-1759.

Farr, T.G., Rosen, P.A., Caro, E., Crippen, R., Duren, R., Hensley, S., Kobrick, M., Paller, M., Rodriguez, E., Roth, L., Seal, D., Shaffer, S., Shimada, J., Umland, J., Werner, M., Oskin, M., Burbank, D., and Alsdorf, D.E., (2007). The shuttle radar topography mission: Reviews of Geophysics, v. 45, no. 2, RG2004.

Guo, H., Li, W., & Wu, J. (2020). Ambient PM2.5 and annual lung cancer incidence: A nationwide study in 295 Chinese counties. International Journal of Environmental Research and Public Health, 17(5), 1481.

Gupta, P., Zhan, S., Mishra, V., Aekakkararungroj, A., Markert, A., Paibong, S., Chishtie, F. (2021). Machine Learning Algorithm for Estimating Surface PM2.5 in Thailand. Aerosol Air Qual. Res. 21, 210105.

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., & Moore, R. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. Remote Sensing of Environment.

Hammer, M. S., van Donkelaar, A., Martin, R. V., Lyapustin, A., Sayer, A. M., Hsu, N. C., ... & Apte, J. S. (2020). Global estimates and long-term trends of fine particulate matter concentrations (1998-2018). Environmental Science & Technology, 54(13).

Hammer, M. S., van Donkelaar, A., Li, C., Lyapustin, A., Sayer, A. M., Hsu, N. C., Levy, R. C., Garay, M. J., Kalashnikova, O. V., Kahn, R. A., Brauer, M., Apte, J. S., Henze, D. K., Zhang, L., & Martin, R. V. (2022). Global Annual PM2.5 Grids from MODIS, MISR and SeaWiFS Aerosol Optical Depth (AOD), 1998-2019, V4.GL.03. Palisades, New York: NASA Socioeconomic Data and Applications Center (SEDAC).

James Moran, Chirada NaSuwan, Ora-Orn Poocharoen, (2019). The haze problem in Northern Thailand and policies to combat it: A review, Environmental Science & Policy: Volume 97.

Li, T., Shen, H., Yuan, Q., Zhang, X., Gu, X., & Huang, C. (2017). Satellite-based estimates of long-term exposure to fine particulate matter and its association with mortality in China. Environmental Science & Technology, 51(10), 1027-1036.

Liu, Y., Liang, L., Li, J., & Zhao, W. (2018). Estimation of PM2.5 concentrations from MODIS satellite data: Validation and application over Beijing, China. Remote Sensing, 10(3), 432.

Loomis, D., et al. (2013). The carcinogenicity of outdoor air pollution. The Lancet Oncology, 14(13), 1262-1263.

Lyapustin, A., Wang, Y. (2022). MODIS/Terra+Aqua Land Aerosol Optical Depth Daily L2G Global 1km SIN Grid V061 [Data set]. NASA EOSDIS Land Processes Distributed Active Archive Center. Accessed 2024-04-22 from https://doi.org/10.5067/MODIS/MCD19A2.061

OpenAI. (2023). ChatGPT (Mar 14 version) [Large language model]. https://chat.openai.com/chat

Pongpiachan, S., Paowa, T., (2015). Hospital out-and-in-patients as functions of trace gaseous species and other meteorological parameters in Chiang-Mai, Thailand. Aerosol Air Qual. Res. 15 (2), 479–493.

Pope, C. A., III, Dockery, D. W., Spengler, J. D., & Raizenne, M. E. (1991). Respiratory health and PM10 pollution. A daily time series analysis. American Review of Respiratory Disease, 144(3 Pt 1), 668-674.

Pope, C. A., III, Burnett, R. T., Thun, M. J., Calle, E. E., Krewski, D., Ito, K., & Thurston, G. D. (2002). Lung cancer, cardiopulmonary mortality, and long-term exposure to fine particulate air pollution. JAMA, 287(9), 1132-1141.

Trang, N., Tripathi, N., (2014). Spatial correlation analysis between particulate matter 10 (PM10) hazard and respiratory diseases in Chiang Mai Province, Thailand. ISPRS - Int. Arch. Photogramm. Remote. Sens. Spat. Inf. Sci. XL-8, 185–191.

Turner, M. C., Krewski, D., Pope, C. A., III, Chen, Y., Gapstur, S. M., & Thun, M. J. (2011). Long-term ambient fine particulate matter air pollution and lung cancer in a large cohort of never-smokers. American Journal of Respiratory and Critical Care Medicine, 184(12), 1374-1381.

United States Environmental Protection Agency (EPA). (1997). National Ambient Air Quality Standards (NAAQS) for Particulate Matter.

van Donkelaar, A., Martin, R. V., Brauer, M., Kahn, R., Levy, R., Verduzco, C., & Villeneuve, P. J. (2010). Global estimates of ambient fine particulate matter concentrations from satellite-based aerosol optical depth: Development and application. Environmental Health Perspectives, 118(6), 847-855.

World Health Organization. (1987). Air quality guidelines for Europe. WHO Regional Office for Europe.

World Health Organization. (2006). Air quality guidelines for particulate matter, ozone, nitrogen dioxide and sulfur dioxide: Global update 2005. WHO Regional Office for Europe.

APPENDIX A

Results of PM2.5 estimation and its impact on lung cancer mortality

According to AOD's limitations, aerosols in the atmosphere may not have the same direct health effects on humans as PM2.5, thus making them less suitable for direct interpretation as an indicator of air quality. However, compared to ground-based air quality measuring stations, using AOD has the advantage of covering greater areas and having longer historical data, and numerous prior research studies discovered an association between AOD and PM2.5. As a result, this study estimates PM2.5 using AOD, following the findings of Liu et al. (2018) and other previous literatures before examining the relationship between these estimated PM2.5 and the mortality from lung cancer.

PM2.5 estimation

Ground-based stations for PM2.5 have limited coverage, especially in developing countries like Thailand, where they cover only 6.6% of the country's 928 districts. Recent research discussed in review literature attempted to estimate PM2.5 using Aerosol Optical Depth (AOD) and found a high correlation. This study also aims to estimate PM2.5 from AOD for Thailand. The estimation of PM2.5 is given by the following equation:

$$PM2.5_{dt} = \beta_0 + \beta_1 AOD_{dt} + \beta_2 Elevation_d + \beta_3 Month_t + \gamma Provinces_d + \varepsilon_{dt} \quad \text{(A1)}$$

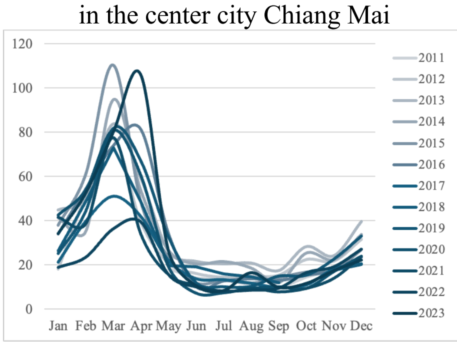
Where $PM2.5_{dt}$ represents particulate matter 2.5 microns or less in diameter from ground-based air quality measuring stations in district d during month t. AOD_{dt} is aerosol optical depth in district d during month t. $Month_d$ is a set of dummy variables for month t. $Elevation_d$ is an average elevation for district d. $Provinces_d$ is a set of dummy variables for each province for district d. To account for unobserved differences across provinces or regions that could impact PM2.5 levels, province or region fixed effects were included in the model. The regression results are presented in Table A1, while examples of PM2.5 forecasting from AOD using the A3 model are depicted in Figures A2 and A4.

Table A1: PM2.5 estimation

Y = PM2.5	(A1)	(A2)	(A3)
AOD	4.709***	-0.014	-0.527
	-0.186	-0.533	-0.525
AOD^2		0.396***	0.426***
		-0.058	-0.057
Elevation		-0.006***	-0.013***
		-0.001	-0.002
Control variables			
12 months		✓	✓
5 Regions		✓	
77 Provinces			✓
Observations	4897	4897	4897
R-squared	0.332	0.661	0.688

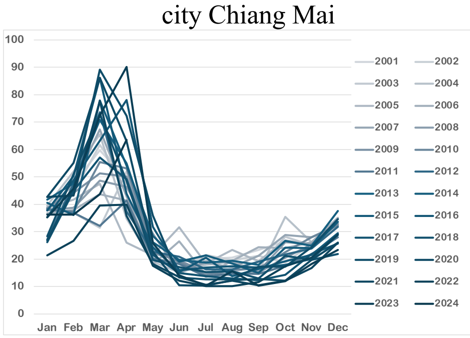
Source: Authors' calculations

Figure A1: PM 2.5 from ground station (36T) in the center city Chiang Mai



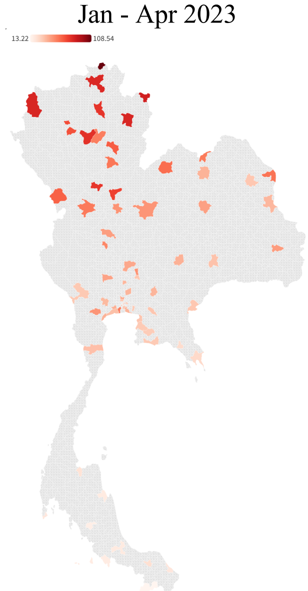
Source: Pollution Control Department, Thailand

Figure A2: Estimated PM 2.5 in the center city Chiang Mai



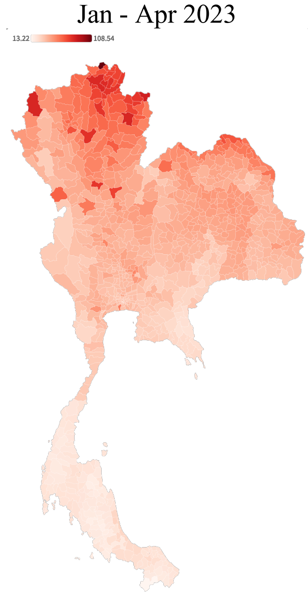
Source: Author's calculation

Figure A3: PM 2.5 from ground station Jan - Apr 2023



Source: Pollution Control Department, Thailand

Figure A4: The estimated average PM 2.5 Jan - Apr 2023



Source: Author's calculation

The impact of estimated PM2.5 on lung cancer mortality

On average estimated PM2.5 correlated with the mortality rate from lung cancer for 20- 10- and 5- years exposure windows. For every one microgram per cubic meter air increase in estimated PM2.5, or µg/m3, the probability of lung cancer mortality rate increases by around 0.3 – 0.4 percentage point for all exposure periods. Furthermore, we discovered that the likelihood of lung cancer mortality increased with every additional unit of estimated PM2.5 during hazy seasons when we separated the PM2.5.

Table A2: The impact of estimated PM2.5 for 20- 10- and 5-years exposure on lung cancer mortality

Y = Average Lung cancer Mortality Rate (2017-2021)	Exposure est PM2.5 on average 20 Years (2001-2021)				Exposure est PM2.5 on average 10 Years (2011-2021)				Exposure est PM2.5 on average 5 Years (2017-2021)			
	(21)	(22)	(23)	(24)	(25)	(26)	(27)	(28)	(29)	(30)	(31)	(32)
PM2.5 (Jan-Dec)	0.575***	0.401***			0.572***	0.318***			0.589***	0.406***		
	-0.046	-0.05			-0.047	-0.056			-0.048	-0.073		
PM2.5 outside haze season (May-Dec)			0.224***	0.191***			0.296***	0.168***			0.332***	0.355***
			-0.055	-0.054			-0.057	-0.058			-0.058	-0.071
PM2.5 Haze (During Jan-Apr different from May-Dec)			0.745***	0.697***			0.557***	0.475***			0.529***	0.410***
			-0.075	-0.085			-0.068	-0.079			-0.069	-0.114
Control variables												
Tobacco consumption rate		✓		✓		✓		✓		✓		✓
Household income		✓		✓		✓		✓		✓		✓
Air pollutants										✓		✓
Observations	928	928	928	928	928	928	928	928	928	928	928	928
R-squared	0.129	0.187	0.198	0.23	0.126	0.188	0.165	0.211	0.127	0.242	0.165	0.251

Note: The analysis covers three periods: a 20-year average (2001-2021), a 10-year average (2011-2021) and a 5-year average (2017-2021). Columns (21) and (22) focus on the 20-year average exposure to PM2.5 for the full year (Jan-Dec), columns (25) and (26) focus on the 10-year average, and columns (29) and (30) focus on the 5-year average. Columns (23) and (24) examine the 20-year average PM2.5 concentration during the non-haze season (May-Dec) and the haze season (Jan-Apr), columns (27) and (28) do the same for the 10-year average, and columns (31) and (32) for the 5-year average. Control variables include tobacco consumption rates, household income, and other air pollutants. Each column presents coefficient estimates, standard errors, number of observations, and R-squared values.
Source: Author's calculation